

UNIT-3

1. Implement the following Boolean functions .

$$F = (\bar{A}+B).(A+\bar{C}).(AB+D) \text{ using NAND gate .}$$

$$F = \overline{ABC} + \overline{ABC} + \overline{ABC} + ABC.$$

2. Design the circuit to realize the Boolean functions:

(a) $Y = \overline{AB} + \overline{AC} + AB.$

(B) $F = ABCD + \overline{A}BCD + ABC\overline{D} + \overline{A}BC\overline{D} + ABCD + \overline{A}BCD + \overline{A}BC\overline{D} + \overline{A}BC\overline{D}$

3. i. what are the characteristics of logic gate
ii. explain XOR gate.
iii. Explain implementation of NOT, AND, OR gates by NOR gates
4. What is combinational logic with example? What are its characteristics? explain the analysis procedure of combinational logic.
- 5.
6. Implement the following Boolean functions using NOR gate:
 - a. $F = (A+\overline{B}+C)(A+\overline{B}+\overline{C})(A+\overline{B}+C)(\overline{A}+\overline{B}+C)$
 - b. $F = \overline{ABC} + \overline{ABC} + \overline{ABC} + ABC$
 - c. What do you mean by universal gate?
 - d. Explain NAND gate.
 - e. Use NAND gates only to solve full adder
 - f. Explain concept of multilevel NAND and NOR gates
 - g. Prove that NAND AND NOR are universal gates
 - h. Design 3 variable and ,nor
 - i. Make T.T and explain NAND, XOR, NOT
 - j. Make T.T for 3 variables nd explain: OR, AND, NOR
 - k. Make circuit diagram: $ab(\overline{c}d + c\overline{d}) + (\overline{a}bc)$
 - l. Design XNOR AND XOR gates
 - m. What is don't care condition
 - n. Design $X = cd(\overline{a}b + ab) + PQ(PQ + \overline{P}\overline{Q})$
 - o. Solve using T.T : $\overline{a.b.c} = \overline{a} + \overline{b} + \overline{c}$
 $(a+b)+c = a+(b=c)$